

Nikhilesh Vijay Chirutala

Software Engineer | Python | AWS | Platform Engineering

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SUMMARY

Software Engineer with 3+ years of experience building and maintaining production applications across cloud, contact center, AI, CRM, and trading platforms. Strong hands-on experience with Python, FastAPI, REST APIs, AWS, SQL, MongoDB, Git, GitHub Actions, and CI/CD. Experienced in application development, backend engineering, distributed service integrations, data processing, cloud deployments, automation, authentication, monitoring, troubleshooting, and production support. Hands-on experience with multi-cluster deployments, failure recovery, operational stability, and reliable cloud services. Strong understanding of software development, testing, SDLC, Agile practices, application resiliency, secure software engineering, system design, and cloud-native platform engineering.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (ES6+), TypeScript

Software Engineering: Software Development, OOP, Data Structures & Algorithms, System Design, Application Development, REST API Design, Backend Development, Microservices, Distributed Systems, Service-to-Service Communication, Troubleshooting, Debugging

Cloud & Platform Engineering: AWS, Amazon S3, Terraform, Kubernetes, Docker, Infrastructure as Code, Container Orchestration, Cloud Application Development, Cloud Deployments, Production Cloud Services, Multi-Cluster Deployments, Linux

Backend & APIs: Python, FastAPI, REST APIs, API Development, Backend Automation, Business Logic, Data Processing, API Integration, Service Integration

DevOps & CI/CD: Git, GitHub, GitHub Actions, CI/CD, SDLC, Continuous Integration, Continuous Deployment, Infrastructure as Code, Terraform, Containerization, Kubernetes, Docker, Feature Flags, Staged Rollouts, Production Deployment, Automation

Application Reliability: Application Resiliency, Operational Stability, Failure Recovery, Retry Strategies, Fault Handling, Idempotent Operations, Validation, Production Support, Incident Resolution, Root Cause Analysis

Data & Analytics: SQL, Data Processing, ETL, Data Validation, Data Analysis, Data Visualization, Reporting, Data-Driven Applications

Observability: Datadog, Grafana, Sentry, Dashboards, Production Monitoring, Application Monitoring, Error Tracking, Service Health Monitoring

Security: Secure Software Development, Authentication, Authorization, RBAC, ACL, JWT, Access Control, Application Security

SDLC & Development Practices: Agile, SDLC, CI/CD, Testing, Debugging, Production Support, Code Quality, Code Review

Databases: MySQL, MongoDB, PostgreSQL, SQL, Database Design

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering, AI-Enabled Applications, Contextual Information Retrieval

Engineering Tools: GitHub, GitHub Actions, Terraform, Kubernetes, Docker, Grafana, SonarQube, Postman, Git

Frontend: React.js, Next.js, Angular, TypeScript, JavaScript, Jinja2

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Develop production-grade Python and FastAPI backend services for a high-volume cloud contact center platform, supporting application workflows, data processing, storage operations, service integrations, automation, and cloud-based operations.
- Migrated a legacy Perl call-recording workflow to a long-running Python service deployed across 40 production clusters, using feature flags, staged rollouts, validation, rollback procedures, monitoring, and production troubleshooting to support reliable releases.
- Designed service-to-service communication for a Python Call Intelligence microservice using APIs, eligibility checks, account and campaign controls, daily processing limits, retry strategies, response validation, and persistence.
- Implemented validation and recovery workflows for incomplete, failed, or inconsistent processing and storage states, improving application resiliency and maintaining reliable downstream operations.
- Supported secure and stable production deployments across multiple service clusters, investigating non-routine technical issues, performing root-cause analysis, and implementing fixes and enhancements.
- Built AWS S3 audit and recovery workflows using deterministic object handling and idempotent re-upload operations to maintain consistency between application records and cloud storage.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry to monitor service health, processing outcomes, failures, and operational issues.
- Integrated SonarQube code-quality checks into the development workflow, ensuring pull requests passed defined quality gates before being merged and promoting consistent coding hygiene across production services.
- Troubleshoot production issues across Python services, REST APIs, databases, cloud storage, and distributed workflows while validating changes before release and supporting operational stability.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog, Grafana, Sentry

- Developed backend services supporting lead routing, call recording management, data processing, cloud storage workflows, and contact center operations.

- Migrated a legacy Perl recording workflow to a long-running Python service deployed across 40 production clusters with feature flags, staged rollout, validation, rollback, and production monitoring.
- Integrated a Python Call Intelligence microservice through service-to-service APIs, implementing eligibility validation, account and campaign controls, daily processing limits, retries, response validation, and persistence.
- Designed validation and recovery workflows for incomplete, failed, and inconsistent recording states to improve application resiliency and operational stability.
- Implemented AWS S3 audit and recovery automation using deterministic object handling and idempotent re-upload operations to maintain reliable cloud storage workflows.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry to monitor processing outcomes, application failures, and service health.
- Investigated production failures across backend services, databases, APIs, and cloud storage, performing root-cause analysis and implementing corrective fixes.

Software Engineer • Mantra Technologies

Nov 2023 – Mar 2026 | Hyderabad, India

- Worked across four production applications, delivering backend, frontend, API integration, business logic, and application features across the software development lifecycle.
- Built backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for application-specific workflows and production applications.
- Developed frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with backend APIs.
- Implemented authentication, RBAC, and ACL-based authorization to enforce secure access controls across application resources.
- Integrated third-party services including Twilio and GitHub API and supported cloud deployments across AWS and Vercel.
- Maintained GitHub Actions CI/CD workflows and handled testing, debugging, bug fixes, code quality improvements, and production support.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, filtering, and back-testing.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated the RAG chatbot into application workflows, enabling conversational access to relevant financial information.
- Built FastAPI/Python backend APIs and business logic for financial data processing, application workflows, and data-driven functionality.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested REST APIs and application workflows with Postman and resolved bugs across CRM functionality and production workflows.

PERSONAL PROJECTS

Cloud-Native Platform Engineering Lab — Cloud-Native Python Platform

Python, FastAPI, Docker, Kubernetes, Terraform, AWS, GitHub Actions, PostgreSQL

- Designed and developed a cloud-native Python/FastAPI application with PostgreSQL, focusing on scalable backend services and production-oriented application architecture.
- Containerized the application using Docker and deployed the services on Kubernetes with health checks, configuration, resource management, and service orchestration.
- Provisioned AWS infrastructure using Terraform Infrastructure as Code, creating repeatable and version-controlled infrastructure and deployment configurations.
- Built GitHub Actions CI/CD workflows to automate testing, container builds, and application deployment across environments.
- Implemented application resiliency through health validation, retry handling, failure recovery, and controlled deployment workflows.

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

EDUCATION

B.Tech — Electrical & Electronics Engineering
Srinivasa Ramanujan Institute of Technology, Anantapur

2019 – 2023 | GPA: 7.73 / 10